**PROGRAM – 2 Write a function to measure distance between 2 given points as (x1,y1) and (x2,y2). Recursively call the function to measure length of 3 sides of triangle ABC, taking 3 co-ordinates as input. Measure the area of triangle ABC. Find out whether given point (x, y) lies inside or outside the triangle.**

**Code :-**

```
import math
def distance(p1, p2, count=1):
    if count == 1:
        return math.sqrt((p1[0] - p2[0])**2 + (p1[1] - p2[1])**2)
    else:
        return distance(p1, p2, count-1)
def area_of_triangle(a, b, c):
    side_ab = distance(a, b)
    side_bc = distance(b, c)
    side_ca = distance(c, a)
    s = (side_ab + side_bc + side_ca) / 2
    return math.sqrt(s * (s - side_ab) * (s - side_bc) * (s - side_ca))
def is_point_inside_triangle(p, a, b, c):
    total_area = area_of_triangle(a, b, c)
    area1 = area_of_triangle(p, b, c)
    area2 = area_of_triangle(a, p, c)
    area3 = area_of_triangle(a, b, p)
    return math.isclose(total_area, area1 + area2 + area3, rel_tol=1e-9)
a = tuple(map(float, input("Enter coordinates of A (x1 y1): ").split()))
b = tuple(map(float, input("Enter coordinates of B (x2 y2): ").split()))
c = tuple(map(float, input("Enter coordinates of C (x3 y3): ").split()))
p = tuple(map(float, input("Enter coordinates of point P (x y): ").split()))
side_ab = distance(a, b)
side_bc = distance(b, c)
side_ca = distance(c, a)
area = area_of_triangle(a, b, c)
inside = is_point_inside_triangle(p, a, b, c)
print(f"Length of AB: {side_ab:.2f}")
print(f"Length of BC: {side_bc:.2f}")
print(f"Length of CA: {side_ca:.2f}")
print(f"Area of triangle ABC: {area:.2f}")
print(f"Is point {p} inside triangle? {inside}")
```

**Output :-**

```
t/WindowsApps/python3.11.exe "c:/Users/ndevr/OneDrive/Desktop/5 DAY Placement workshop/DAY 3/Problem
1/sol2.py"
Enter coordinates of A (x1 y1): 0 0
Enter coordinates of B (x2 y2): 4 0
Enter coordinates of C (x3 y3): 0 3
Enter coordinates of point P (x y): 2 1
Length of AB: 4.00
Length of BC: 5.00
Length of CA: 3.00
Area of triangle ABC: 6.00
Is point (2.0, 1.0) inside triangle? True
PS C:\Users\ndevr\OneDrive\Desktop\5 DAY Placement workshop>
```